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DEVICE CONTROL Through A PC's USB Port Using Visual Basic

ADEEB RAZA

This Windows-based program, created with Microsoft Visual Basic (VB6), features an intuitive user graphic interface (UGI) and operates seamlessly on the Windows platform. It interfaces with a personal computer's (PC's) USB port via a hardware module, which includes a microprocessor-based electronic circuit capable of controlling up to eight appliances through the PC.

The program transmits specific instructions to the microprocessor-based interface card via the USB port. These instructions are pre-programmed and stored in the microprocessor's memory. Actions are executed through a designated relay number linked to an input/output (I/O) pin of the microprocessor and driver circuit.

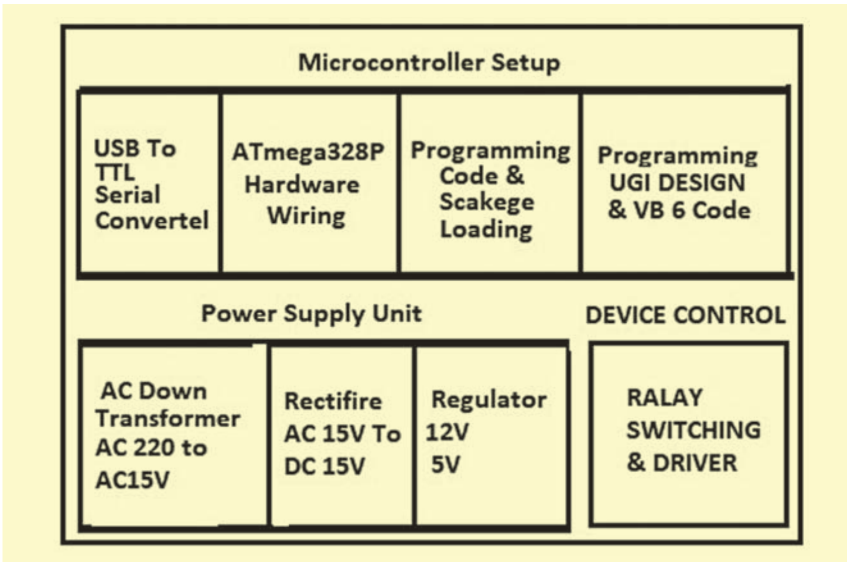

Fig. 1: Block diagram of the project

At its core is the ATmega328P, a single-chip, 8-bit microcontroller housed in a 28-pin DIL package. This microcontroller is both cost-effective and widely available in the Indian market. As it is the same IC used in Arduino Uno, the ATmega328P circuit can be easily replaced with an Arduino Uno board, if preferred.

The Arduino IDE, required to program the ATmega328P, is available for free download at *https://www.arduino.cc/en/software*. A block diagram illustrating the program's functionality is shown in Fig. 1, while the complete bill of materials is provided in the table on this page.

This program involves using Visual Basic (VB) to communicate with an external hardware module through the USB port. A basic understanding of VB

BILL OF MATERIALS

Components	Quantity
220V primary to 15V, 1A secondary transformer (T1)	1
Rectifier diode 1N4007 (D1-D12)	12
Capacitor 1000µF, 25V (C1)	1
12V voltage regulator IC 7812 (IC1)	1
5V voltage regulator IC 7805 (IC4)	1
Relay 12V SPDT (RLY 1- RLY8)	8
Relay driver IC ULN2803A (IC2)	1
5mm red LED for power indication	1
1k resistor (R1)	1
Microcontroller ATmega328P (IC3)	1
28-pin DIL IC base (optional for breadboard)	1
18-pin DIL IC base (optional for breadboard)	1
Crystal 16MHz (X1)	1
22pF ceramic capacitor	2
USB-to-TTL serial converter (see Fig. 2)	1
Jumper wires	As required
Relay connectors	8

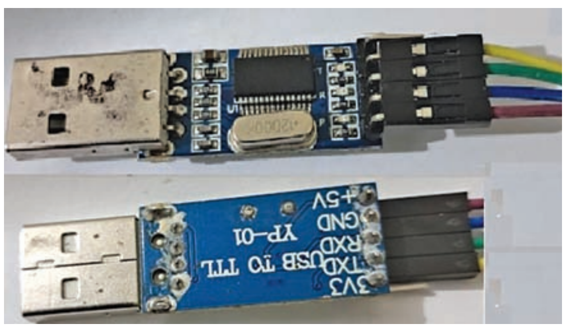

Fig. 2: USB-to-TTL serial converter